AI-Driven Skill Proof System

Domain :

AI | EdTech | Skill Evaluation Systems

Problem Statement :

Design and develop an AI-Driven Skill Proof System that evaluates and verifies a student’s abilities based on their actual performance and behavior while solving tasks or problems.

The system should attempt to answer the question:

“Can this person actually do what they claim?”

Background :

Traditional methods of evaluating skills such as certificates, resumes, and project submissions often fail to accurately reflect a person’s true capabilities. Many individuals can claim expertise without demonstrating genuine understanding or practical ability.

As a result, industries and organizations face difficulty distinguishing between genuine talent and superficial claims of expertise.

A more reliable approach is to evaluate individuals based on real actions, decision-making patterns, and problem-solving performance rather than static credentials.

Objective

Create a prototype system that continuously analyzes a student’s performance while solving tasks and produces transparent, interpretable skill scores reflecting their real abilities.
The system should provide a clear skill profile that helps institutions or employers understand a student's strengths and weaknesses.

Key Features :

1. Task or Challenge Platform

The system should provide tasks that allow students to demonstrate their skills.
Examples include:

- Coding challenges
- Logical problem-solving tasks
- Conceptual questions

2. Behavior & Performance Analysis

The platform should track and analyze factors such as:

- Correctness of solutions
- Time taken to solve tasks
- Number of attempts
- Error patterns
- Problem-solving approach

These metrics act as evidence of actual skill.

3. Skill Confidence Scoring

Based on the collected evidence, the system should generate skill confidence scores.
Example output:

Skill	Confidence Score
Programming	82%
Problem Solving	74%
Debugging	68%

4. Skill Profile Dashboard

The platform should present results in a clear visual dashboard displaying:

- Skill scores
- Performance insights
- Strengths and areas for improvement

5. Transparency & Explainability

The system should provide explanations for how scores were generated, such as:

- Number of successful attempts
- Complexity of tasks solved
- Consistency of performance

Example Use Case

A student attempts several programming challenges on the platform.
The system records performance data such as:

- Solution accuracy
- Code efficiency
- Time taken
- Number of debugging attempts

Using this information, the platform generates a verified skill profile that reflects the student’s actual capabilities.

Expected Deliverables

Teams should present a working prototype that includes:

- A task or challenge interface
- A system for collecting performance metrics
- A skill scoring mechanism
- A dashboard displaying skill profiles and insights

Suggested Technologies

Participants may use any technology stack. Suggested tools include:

Programming Languages

- Python
- JavaScript

AI / Data Analysis

- Machine learning models
- LLM APIs
- Data analytics libraries

Frameworks

- Flask / FastAPI
- Node.js

Frontend

- React
- Streamlit

Outcome

Participants will build a prototype system demonstrating how skills can be verified through real performance and behavioral analysis, enabling a more accurate and transparent approach to evaluating talent.